

SymPy Bug Card

Bug 39: Limit of Bessel quotient derivative is negative infinity

Task. Evaluate the limit of the derivative at the origin:

$$\lim_{x \rightarrow 0} \frac{d}{dx} \left(\frac{J_1(x)}{x} \right).$$

SymPy's answer.

$$\text{limit}\left(\frac{d}{dx} \frac{\text{besselj}(1, x)}{x}, x, 0\right) = -\infty.$$

Correct answer.

$$0.$$

Explanation. Since

$$J_1(x)/x = \frac{1}{2} - \frac{x^2}{16} + O(x^4),$$

its derivative is $-x/8 + O(x^3)$, which tends to 0. The infinite answer is therefore not an equivalent form or branch convention.

Likely root cause. The diagnosis traces the wrong result to `Add._eval_as_leading_term` in `sympy/core/add.py`: after the $1/(2x)$ and $-1/(2x)$ leading terms cancel, the fallback asks for too low an `_eval_nseries` order and accepts a spurious negative-power term. This is a new failure mode with no public precedent.

Artifact (GitHub). [bug-report/bug-039-limit-of-bessel-quotient-derivative-is-negative-infinity/](#)